MIL-R-12405D(ER)
5 May 1982
SUPERSEDING
MIL-R-12405C(EL)
1 November 1967

MILITARY SPECIFICATION

RECORDER, WIND DIRECTION AND SPEED RO-2( )/GMQ

This specification is approved for use by Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of meteorological equipment, designated as Recorder, Wind Direction and Speed, RO-2( )/GMQ (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Communication Equipment.
MIL-M-13231	- Marking of Electronic Items.
MIL-F-14072	- Finishes for Ground Signal Equipment.
MIL-C-45662	- Calibration of Standards.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, Combat Surveillance and Target Acquisition Laboratory, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6660

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-169	- Extreme Temperature Cycle.
MIL-STD-170	- Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Military Standard Environmental Test Methods for Aerospace and Ground Equipment.

2.1.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ELECTRONICS COMMAND

SC-DL-168602	- Recorder, Wind Direction and Speed RO-2( )/GMQ.
SPI 1G00236	- Special Packaging Instruction.

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.2g).

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and SC-DL-168602. The requirements of MIL-P-11268 shall apply when the part, material or process is not specified by drawing.

3.3 Recycled, virgin and reclaimed materials. Without jeopardizing the intended use of equipment furnished to this specification, selection of materials will be governed by the following:

a. There is no exclusion to the use of recovered materials.
b. There is no requirement that equipment be manufactured from virgin materials.
c. Reclaimed materials are required to the maximum extent possible.

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3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and SC-DL-168602.

3.5 Marking.

3.5.1 General. Marking shall conform to MIL-M-13231 and SC-DL-168602.

3.5.2 Serial numbers. Each Recorder, Wind Direction and Speed RO-2( )/GMQ shall have a serial number.

3.6 Interchangeability. Like units, assemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.10).

3.7 Tropicalization of material.

3.7.1 Fungus. Equipment and assemblies shall show no evidence of viable fungus or corrosion when subjected to 28 days of fungus test in accordance with 4.9.6.1. (Corrosion is defined as any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.)

3.7.2 Salt fog. After testing as specified in 4.9.6.2, the equipment shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface.

3.8 Performance. The equipment shall meet all the requirements of this specification with an input voltage of 115 ±10 VAC at 60 Hz.

3.8.1 Wind direction response.

3.8.1.1 Accuracy (direction). The accuracy of the wind direction indication shall be within ±2 degrees.

3.8.1.2 Speed of response. The speed of response shall not exceed 6.0 seconds per 178 degrees change in wind direction.

3.8.1.3 Recorder pen repositioning time. The time required for the recorder pen mechanism to automatically change position by 360 degrees when the pen reaches either scale limit, shall not exceed 6.0 seconds.

3.8.2 Wind speed response.

3.8.2.1 Wind speed recording voltages. The wind speed recording shall vary linearly with the input recording voltage. A reading of 120 knots shall be recorded with an input of 14.6 ±.05 VDC. A reading of 240 knots shall be recorded with an input of 29.2 ±0.1 VDC.

3.8.2.2 Accuracy (wind speed recording). The wind speed indication shall be accurate to one percent of full scale throughout the 0-120 and the 0-240 knot ranges and the 0-140 and the 0-280 miles-per-hour ranges.

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3.8.2.3 Wind speed response. The time required for the pen to travel from 0 to 120, or 0 to 240 knots (full scale) and record normally, shall not exceed 1.2 seconds.

3.8.2.4 Wind speed sensitivity. The wind speed recording circuits shall be capable of recording a wind speed change of one percent in any of its ranges.

3.8.2.5 Wind speed meter calibration. The wind speed meter shall be linearly calibrated so that 14.6 ±.05 dc volts input indicates a wind speed of 120 knots.

3.8.2.6 Wind speed meter accuracy. The wind speed meter shall have an accuracy within ±1.0% of full scale value up to either 120 knots or 140 miles per hour, and 240 knots or 280 miles per hour.

3.8.3 Chart drive.

3.8.3.1 Chart speed. The chart shall travel at a nominal rate of 3 inches (76.2mm) per hour under all specified conditions. It shall also operate at a speed of 6 inches (152.4mm) per hour by a change of gears.

3.8.3.2 Chart drive accuracy. The chart drive mechanism shall drive the chart with an accuracy of ±2 minutes per 24 hours.

3.8.3.3 Friction clutch. The friction clutch shall allow manual repositioning of the chart without interference with, or disconnection of the normal electrical chart drive.

3.8.4 Pen tracings.

3.8.4.1 Pen recording lines. The pens shall draw lines 0.012 ±.004 in (0.3 ±0.1mm) wide.

3.8.4.2 Vane trace. The chart record shall at all times present only one trace of the vane movement. The wind direction recordings shall not be obscured by pen operations across the chart, caused by yawing of the transmitter about the direction repositioning points of the recorder.

3.9 Service conditions. The equipment shall be submited to and meet the service conditions whereby specified performance is defined as meeting the following requirements:

Inspection	Requirement paragraph	Test method paragraph
Wind direction response	3.8.1	4.8.1
Wind speed response	3.8.2	4.8.2
Chart drive	3.8.3	4.8.3
Pen tracings	3.8.4	4.8.4

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3.9.1 Temperature (operating) (see 4.9.1). The equipment shall be operable at temperatures ranging from 32°F to 125°F (0°C to 51.6°C). (The higher temperature includes sun-load effect.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours for any continuous period of time.

3.9.2 Temperature (non-operating) (see 4.9.1). The equipment shall be exposed to temperatures ranging from -80°F to 160°F (-62.2°C to 71°C). Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours for any continuous period of time.

3.9.3 Relative humidity. The equipment shall be subjected to the tests specified in 4.9.2 and shall meet the performance requirements specified therein.

3.9.4 Elevation. (see 4.9.3)

a. Operating. Up to 10,000 feet (3,048m) above sea level.

b. Nonoperating. Up to 50,000 feet (15,240m) above sea level.

3.9.5 Sand and dust. The equipment shall be subjected to the tests specified in 4.9.4 and shall meet the performance requirements specified therein and show no evidence of deterioration of either external or internal parts.

3.9.6 Bench-handling. The equipment shall meet full specification performance with no physical damage except surface abrasion after being subjected to the tests of 4.9.5.

3.10 Preconditioning. The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.

3.11 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the following applicable portions of MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling:
  Slack.
  Protection.
  Clearance.
  Splicing.
  Connections, general.
  Grounding, general
  Shielding on wire and cable.
Soldering and brazing.
  Cleaning of units.
  General requirements for securing of parts.
Process for welding.
  Cleaning prior to welding.
  Cleaning after welding.
Controls.
Self-tapping screws.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging) (see 4.3).
b. Inspections covered by subsidiary documents (see 4.4).
c. Quality conformance inspection.
   (1) Quality conformance inspection of equipment before packaging (see 4.5).
   (2) Quality conformance inspection for packaging (see 4.11).

4.3 First article inspection. This inspection will be performed by the contractor unless otherwise specified in the contract. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4 and the inspection specified for group A, group B and group C testing.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirements specified in the contract:

Item	Requirement paragraph
Finish	3.4
Marking	3.5

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. This inspection, including sampling, shall conform to table I and the ordinary inspection procedures of MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the government.

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TABLE I. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)	
			Major	Minor
Visual-mechanical	3.11	4.7		
Recorder unit (with chart drive mechanism removed)	3.11	4.7	1.5	6.5
Chart drive mechanism	3.11	4.7	1.0	4.0
Complete recorder unit with case	3.11	4.7	1.0	4.0

TABLE II. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (% defective)
Performance			
Wind direction response	3.8.1	4.8.	
Accuracy	3.8.1.1	4.8.1.1	
Speed of response	3.8.1.2	4.8.1.2	
Recorder pen repositioning time	3.8.1.3	4.8.1.3	
Wind speed response	3.8.2	4.8.2	
Wind speed recording voltages	3.8.2.1	4.8.2.1	
Accuracy (wind speed recording)	3.8.2.2	4.8.2.2	2.5
Wind speed response	3.8.2.3	4.8.2.3	(combined
Wind speed sensitivity	3.8.2.4	4.8.2.4	for the
Wind speed meter calibration	3.8.2.5	4.8.2.5	group 1/)
Wind speed meter accuracy	3.8.2.6	4.8.2.5	
Chart drive	3.8.3	4.8.3	
Chart speed	3.8.3.1	4.8.3.1	
Chart drive accuracy	3.8.3.2	4.8.3.1	
Friction clutch	3.8.3.3	4.8.3.2	
Pen tracings	3.8.4	4.8.4	
Pen recording lines	3.8.4.1	4.8.4	
Vane trace	3.8.4.2	4.8.4	
Interchangeability	3.6	4.10	6.5

1/ All operational tests are considered to be major.

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4.5.2 Group B inspection. This inspection, including sampling, shall conform to table II and to the procedures for small sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.5.3 Group C inspection. Group C inspection shall consist of the tests specified in table III and shall be performed on sample units that have been subjected to and met groups A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

TABLE III. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Subgroup 1		
Moisture-resistance	3.9.3	4.9.2
Dust	3.9.5	4.9.4
Subgroup 2		
Temperature	3.9.2	4.9.1
Elevation	3.9.4	4.9.3
Bench-handling	3.9.6	4.9.5
Tropicalization	3.7	4.9.6

4.5.3.1 Sampling for inspection. Two items of production for group C inspection shall be selected from the first 10 items produced and subjected to subgroups 1 and 2 inspection and one from each 20 items produced thereafter shall be subjected to subgroup 1 inspection. The items shall be selected without regard to their quality and shall be selected at random from each group of units produced. When first article samples are required by contract, group C inspection is not required for the two production items of the first 10 produced.

4.5.3.2 Noncompliance. The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units of product which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the government) and groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

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4.6 Bounce preconditioning. The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneatales, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a strip of material, 0.03125 in. (0.8mm) thick, under one corner or edge of the unit to a distance of 3 inches (76.2mm) as the unit bounces. The unit shall be subjected to this preconditioning for 5 minutes. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.7 Visual and mechanical inspection. The equipment shall be inspected for compliance with requirement 9 of MIL-STD-454.

4.8 Performance tests. Test equipment shall be calibrated in accordance with this specification. When the calibration is not specified, MIL-C-45662 shall apply. The nominal input voltage of the RO-2( )/GMQ is 115 VAC at 60 Hz.

4.8.1 Wind direction response. Test at the inputs specified by 3.8.

4.8.1.1 Accuracy. The wind direction mechanism shall be rotated through one complete revolution in each direction, and the change in chart recordings checked at 30 degree increments conforming to 3.8.1.1.

4.8.1.2 Speed of response. With the wind direction pen positioned on one side of the chart paper, the power to the equipment "off," and the synchro transmitter rotated 178 degrees, apply the alternating current (ac) power. The time required for the pen to travel to the new position shall not exceed that specified in 3.8.1.2. Repeat with the pen positioned on the opposite side of the recording paper. Perform each test three times; the accuracy of each test shall conform to 3.8.1.2.

4.8.1.3 Recorder pen repositioning time. A wind direction signal shall be applied to the wind direction recording mechanism and continuously varied in such a manner as to cause the pen to automatically change position by 360°. This test shall be repeated with the pen traveling in the opposite direction. Repeat this test three times in each direction. The time equired for each repositioning operation shall be measured for conformance to 3.8.1.3.

4.8.2 Wind speed response.

4.8.2.1 Wind speed recording voltages. The wind speed recording voltages shall conform to 3.8.2.1.

4.8.2.2 Accuracy (wind speed recording). The wind speed mechanism shall be zero adjusted, using the zero adjust range control to align the recording pen with the zero knots line on the recording chart. A wind speed voltage shall be applied to the recording mechanism and gradually increased. Voltage readings shall be taken at each successive ten knots point of the chart, starting at the 10 knot marker. Repeat the test in the reverse direction. This same test shall then be repeated, again in both directions. All four readings shall be compared with each other. The variation in readings at any point shall conform to 3.8.2.2.

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4.8.2.3 Wind speed response. The wind speed pen shall be zero positioned on the chart and the equipment power turned off. The wind speed synchro shall then be positioned to 120 knots, the voltage source set to the equivalent of 120 knots and then, energize the equipment. The time required for the pen to travel to the 120 knot point on the chart shall be recorded. Repeat this test with the synchro positioned to the 240 knot point. Again record the elapsed time. Repeat each test three times; the response time of each test shall conform to 3.8.2.3.

4.8.2.4 Wind speed sensitivity. The wind speed mechanism shall be operated to record a constant wind speed of 10 knots. The wind speed voltages shall then be increased slowly until a noticeable reading change is obtained. The pen shall then be returned to the 10 knot reading and the voltage decreased slowly until again a noticeable change in the reading is obtained. This test shall be repeated at 40, 70, 100, 120, 130, 160, 190, 220 and 240 knots. The results shall meet the requirements of 3.8.2.4.

4.8.2.5 Wind speed meter calibration and accuracy. During the tests of 4.8.2.1, the dial calibration and accuracy shall be checked for conformance to 3.8.2.5 and 3.8.2.6.

4.8.3 Chart drive. Test at the inputs specified in 3.8.

4.8.3.1 Chart drive speed and accuracy. The equipment shall be set up for normal operation (60 Hz power source) and the chart drive mechanism run for 48 hours. The elapsed time shall be periodically compared to the total length of chart run-off. The results shall meet the requirements of 3.8.3.1 and 3.8.3.2.

4.8.3.2 Friction clutch. The chart shall be manually repositioned, a minimum of one full turn of the roller, to ascertain conformance to 3.8.3.3.

4.8.4 Pen tractings and vane trace. All the pen tracings recorded during any of the tests prescribed herein shall be examined for conformance to 3.8.4.1 and 3.8.4.2.

4.9 Service conditions tests.

4.9.1 Temperature test. The equipment shall be subjected to the temperature cycle shown in MIL-STD-169 and meet the full specification requirements in accordance with the following schedule:

Steps	Test method paragraph	Measurement
1, 4, 9, 10	4.8.1.2	Wind direction speed of response.
1, 4, 9, 10	4.8.1.3	Recorder pen respositioning time.
1, 4, 9, 10	4.8.2	Wind speed response.
9	4.8.3.1	Chart drive speed.

4.9.2 Moisture-resistance test. The equipment shall be tested as follows:

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4.9.2.1 Test conditions.

a. Do not move equipment from the humidity chamber for measurements.
b. Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.9.2.2 Test method.

a. Dry at 130° +5°F (54° +2.8°C) for 24 hours.
b. Condition at 77° +5°F (25° +2.8°C) and 40 to 50 percent relative humidity for 24 hours.
c. With a fresh roll of chart paper installed take measurements as specified in 4.9.2.3, and readjust or realign as necessry to meet full specification requirements.
d. Remove the chart roll and subject the equipment to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-170. The equipment shall be energized during the periods shown in the standard to check operability only.
e. After cycling has been completed, condition the equipment for 24 hours at 77° +5°F (25° +2.8°C) and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.9.2.3, using the same chart roll used for measurements in "c" above.

4.9.2.3 Performance. The equipment shall be tested in accordance with the following schedule and the measurements shall meet the full specification requirements:

Paragraph	Measurement
4.8.1.2	Wind direction speed of response.
4.8.1.3	Recorder pen repositioning time.
4.8.2	Wind speed response.
4.8.3.1	Chart drive speed.

4.9.2.4 Failure. If the equipment fails to operate during cycling, it does not pass the test. If the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.9.2.2e, it does not pass the test.

4.9.3 Elevation test. The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

a. With the equipment placed in the chamber, operating and stabilized at room temperature, take the following measurements:

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Test method paragraph	Measurement
4.8.1.2	Wind direction speed of response.
4.8.1.3	Recorder pen repositioning time.
4.8.2	Wind speed response.

b. Reduce the barometric pressure to 20.6 ±0.1 inches (523 ±2.5mm) of mercury (simulated 10,000 feet (3048m) above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.9.3a. The equipment shall meet specified performance for those measurements.

c. Remove the power from the equipment and further reduce the barometric pressure to 3.4 ±0.1 inches (86 ±2.5mm) of mercury (simulated 50,000 feet (15240m) above sea level). Maintain this condition for 1 hour.

d. Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.9.3a. The equipment shall meet specified performance requirements with these measurements.

4.9.4 Sand and dust test. The equipment shall be tested as follows:

a. Place the equipment in its normal operating position, in the test chamber. Raise and maintain the sand and dust density at 0.1 to 0.5 gram per cubic foot within the test space. The relative humidity shall not exceed 30 percent at any time during the test.

b. The sand and dust shall be of angular structure and shall have a particle size such that the particles can pass through screens as follows:

Screen size (US Standard sieve series)	Percent
100-mesh	100
140-mesh	98 ±2
200-mesh	90 ±2
325-mesh	75 ±2

c. Chemical analysis of the sand and dust shall be as follows:

Substance	Percent by weight
Silicon dioxide (SiO_2)	98.0 ±1.0
Ferric oxide (Fe_2O_3)	1.0 ±1.0
Titanium dioxide (TiO_2)	1.0 ±1.0
Aluminum oxide (Al_2O_3)	0.5 ±0.5
Magnesium oxide (MgO)	0.5 ±0.5
Ignition losses	1.0 ±1.0

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d. Maintain internal temperature of test chamber at 77° ±4°F (25° ±2.2°C) for a period of 6 hours, with sand and dust velocity through the test chamber between 100 and 500 feet (30 and 152m) per minute.

e. After 6 hours at above conditions, raise the temperature to 160° ±4°F (71° ±2.2°C) and maintain it at this value for 6 hours. The other conditions shall remain the same.

f. Remove equipment from test chamber and allow it to cool to room temperature. Accumulated dust shall be removed by brushing, wiping, or shaking, with care taken to avoid introduction of additional dust into the equipment. However, dust shall not be removed by either blast or vacuum cleaning. The equipment shall meet specified performance for the following measurements:

Test method paragraph	Measurement
4.8.1.2	Wind direction speed of response.
4.8.1.3	Recorder pen repositioning time.
4.8.2	Wind speed response.
4.8.3.1	Chart drive speed.

g. Open the equipment. There shall be no evidence of deterioration of either external or internal parts.

4.9.5 Shock test: bench-handling. The equipment chassis shall be placed in a suitable position for servicing on a solid, 2-inch (50.8mm) fir bench top. The test shall be performed as follows, in a manner simulating shocks likely to occur during servicing:

a. Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

b. Repeat "a" with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.9.6 Tropicalization.

4.9.6.1 Fungus. The equipment and assemblies shall be subjected to 28 days of the fungus test specified in method 508.1 of MIL-STD-810. For gasket or other non-hermetically sealed equipments, the parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The equipment shall be subjected to the test in this open condition.

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4.9.6.2 Salt fog. The equipment and assemblies shall be tested in their operating configuration in accordance with method 509.1 of MIL-STD-810 for 48 hours. Equipment and assemblies shall not be opened and operation is not required.

4.10 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover, and insertable assemblies, when such dimensions affect mating of parts.
b. Gear size and mesh.
c. Dimensions of cavities, when such dimensions affect insertion of items.
d. Recording arm length and fit.
e. Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, cover, and mountings.
f. Chart size, drive hole size and spacing.
g. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; location of the mating parts on the plug-in assembly or major unit.
h. Size and form of special threads.

4.11 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI 1G00236.

6. NOTES

6.1 Intended use. Recorder, Wind Direction and Speed, RO-2( )/GMQ provides a continuous, visible, permanent and easily interpreted record of wind speed and direction on a single roll of chart paper. This recorder has been designed to record the data transmitted by Wind Measuring Set AN/GMQ-11( ) and will operate correctly with or without Indicator, Wind Direction and Speed, ID-373( )/GMQ-11 in the system.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.
b. Level A or B preservation and packing (see section 5).
c. If rough handling tests are required (see section 4).
d. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.
e. Marking and shipping of samples.
f. Place of final inspection.
g. Number of first articles, if required, to be submitted for approval (see 3.1).

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6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example, RO-2W/GMQ. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-12405D(ER) RECORDER, WIND DIRECTION AND SPEED RO-2( )/GMQ

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

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C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

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